

HDOC - Day 26

Menu-Driven Right-Aligned Patterns

Algorithm

1. Start and display a menu for the three right-aligned patterns and Exit.
2. Read the user choice and use switch-case to call the corresponding parameterless function.
3. Inside the selected function, read a positive number of rows n.
4. Use nested loops: one loop for leading spaces and another for stars/numbers.
5. Display the selected pattern, return to the menu, and repeat until Exit is selected.
6. Stop.

Test Case Table

Test	Choice	n	Expected output
1	1	4	Right-aligned increasing star triangle
2	2	4	Right-aligned increasing number triangle
3	3	4	Right-aligned decreasing star triangle
4	4	-	Exit message

C Program

```
#include <stdio.h>
void incStar(); void incNum(); void decStar();
int main(){
    int ch;
    do{
        printf("\n1. Right-aligned increasing triangle\n");
        printf("2. Right-aligned increasing number triangle\n");
        printf("3. Right-aligned decreasing triangle\n4. Exit\n");
        printf("Enter choice: "); scanf("%d",&ch);
        switch(ch){case 1:incStar();break; case 2:incNum();break;
                   case 3:decStar();break; case 4:printf("Exiting...\n");break;
                   default:printf("Invalid choice\n");}
    }while(ch!=4); return 0;
}
void incStar(){int n,i,j; printf("Enter rows: ");scanf("%d",&n);
for(i=1;i<=n;i++){for(j=1;j<=n-i;j++)printf(" ");
for(j=1;j<=i;j++)printf("* ");printf("\n");}}
void incNum(){int n,i,j; printf("Enter rows: ");scanf("%d",&n);
for(i=1;i<=n;i++){for(j=1;j<=n-i;j++)printf(" ");
for(j=1;j<=i;j++)printf("%d ",j);printf("\n");}}
void decStar(){int n,i,j; printf("Enter rows: ");scanf("%d",&n);
for(i=n;i>=1;i--){for(j=1;j<=n-i;j++)printf(" ");
for(j=1;j<=i;j++)printf("* ");printf("\n");}}
```

Compilation, Execution & Testing

Test Case 1 - Increasing star triangle

```
puppie@Anvikas-MacBook-Air Desktop % clang day26.c -o day26
puppie@Anvikas-MacBook-Air Desktop % ./day26
Enter choice: 1
Enter rows: 4
      *
     * *
    * * *
   * * * *
```

Test Case 2 - Increasing number triangle

```
puppie@Anvikas-MacBook-Air Desktop % ./day26
Enter choice: 2
Enter rows: 4
      1
     1 2
    1 2 3
   1 2 3 4
```

Test Case 3 - Decreasing star triangle

```
puppie@Anvikas-MacBook-Air Desktop % ./day26
Enter choice: 3
Enter rows: 4
* * * *
 * * *
  * *
   *
```

Test Case 4 - Exit

```
puppie@Anvikas-MacBook-Air Desktop % ./day26
Enter choice: 4
Exiting...
puppie@Anvikas-MacBook-Air Desktop %
```

Result: The menu-driven C program for right-aligned patterns was implemented using switch-case, parameterless functions and nested loops, and was tested successfully.